

The Design of Improved Dynamic AES and Hardware Implementation Using FPGA

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Abstract

ByteSub is a key module in the AES hardware architectures. Most of the ByteSub design is accomplished by look-up table method. To reduce the complexity of AES module and improve the performance and security, we propose a novel dynamic ByteSub generator, which is composed of finite field inverse and multipliers. In our design, look-up table method is no more essential to the ByteSub hardware architecture. The proposed design is implemented on Altera APEX20KE FPGA family of devices. It needs 44218 logic cell elements on FPGA chip and the performance can be achieve to 57.2 MHz. Our proposed design does not need any memory elements.

1 Introduction

In 2001, the National Institute of Standards and Technology (NIST) announced Rijndael algorithm to be Federal Information Processing Standard (FIPS) 197 [10] - Advanced Encryption Standard (AES). Hence, the AES replaced the Data Encryption Standard (DES) [11]. By replacing the DES, AES algorithm is the standard of most cryptography applications with software or hardware implementation since then. The goal of our design is to improve the performance and security integrity. In this paper, we proposed the dynamic AES algorithm hardware design by using the standard AES algorithm and finite field inverse algorithm circuit on FPGA devices. The details are described in the following sections.

2 The Advanced Encryption Standard

The AES algorithm is a symmetric block cipher. The length of the plaintext and ciphertext data block is 128 bits. The length of the key block can be 128,

192 or 256 bits. In this paper, the length of AES algorithm proposed is limited to 128-bit data and 128-bit key blocks. The block of data and key in AES algorithm can be presented as a 4×4 matrix of bytes. The encryption/decryption operation in AES is composed of multiple rounds, which is the basic operating unit in AES. Every round consists of four transformations except the final round in AES encryption/decryption operation. The four transformations are described as below:

- ByteSub - a single non-linear transformation, applied to each byte of the data.
- ShiftRow - considering the data as a 4×4 matrix of bytes, reorders the bytes of each row cyclic.
- MixColumn - considering the data as a 4×4 matrix of bytes, a linear transformation applied to columns of the matrix.
- AddRoundKey - XORed with a round key and the intermediate data block, which is produced after the previous three transformations.

The initial step of AES is XORed with original data and key block before any AES round operations. The MixColumn transformation is excluded in the final round operation of AES. The complete encryption/decryption operation of AES algorithm is shown in Figure 1.

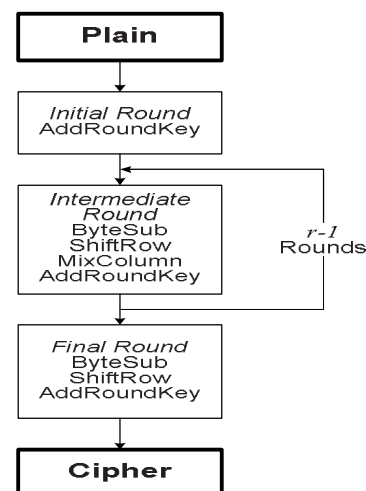


Figure 1. The complete operations of AES.

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A full detailed description of the AES algorithm can be found in the FIPS-197 [10] documents.

3 Improvement of AES

I. The multiplier and inverse in $GF(2^m)$

ByteSub transformation is critical to the performance of AES algorithm. Most of the traditional methods utilize look-up tables (S-Box table) to implement ByteSub transformation [3,8]. A single S-box look-up table required 256 bytes memory elements in it. In general, 8K bytes of look-up table construct the 32 S-Boxes for AES encryption and decryption operations with 128 bits length of data and key block. Hence, this kind of architecture is not suitable for area limited applications, such as the smart card chip design [4]. To avoid the use of look-up table, the finite field multiplier and inverse modules are used in the proposed designed.

Multiplier

The AES algorithm mathematical operation is based on $GF(2^8)$. There are 30 kinds of irreducible polynomials in $GF(2^8)$ [7,10]. The irreducible polynomial used in standard AES is $x^8+x^4+x^3+x+1$, which is presented as $m(x)$.

According to M.H. Jing et al. [5], we developed a novel 8-bit multiplier over $GF(2^8)$ with extra 8-tuples coefficients over $GF(2)$, which can appoint any chosen irreducible polynomial over $GF(2^8)$ by users. The architecture of this novel 8-bit multiplier is shown in Figure 2.

Inverse

Traditionally, the finite field inverse calculation can be achieved by a series of finite field multipliers [1,2,5]. Considering a finite field element over $GF(2^m)$ A , the inversion A^{-1} can be calculated by a series of power of A . According to Stephen B. Wicker [9], the finite field inversion can be expressed as

$$A \bullet A^{-1} = 1 = A^{2^m-1} = A \bullet A^{2^m-2} \quad (1)$$

Thus, the finite field $GF(2^8)$ multiplicative inversion A^{-1} , performed in ByteSub transformation, can be calculated by $A^{2^8-2} = A^{254}$. By using the structure of the systolic array, A^{254} can be represented by the multiplications of the square property of A and A^2 [9]. The block diagram of A^{-1} circuit, which uses square property, is shown in Figure 3.

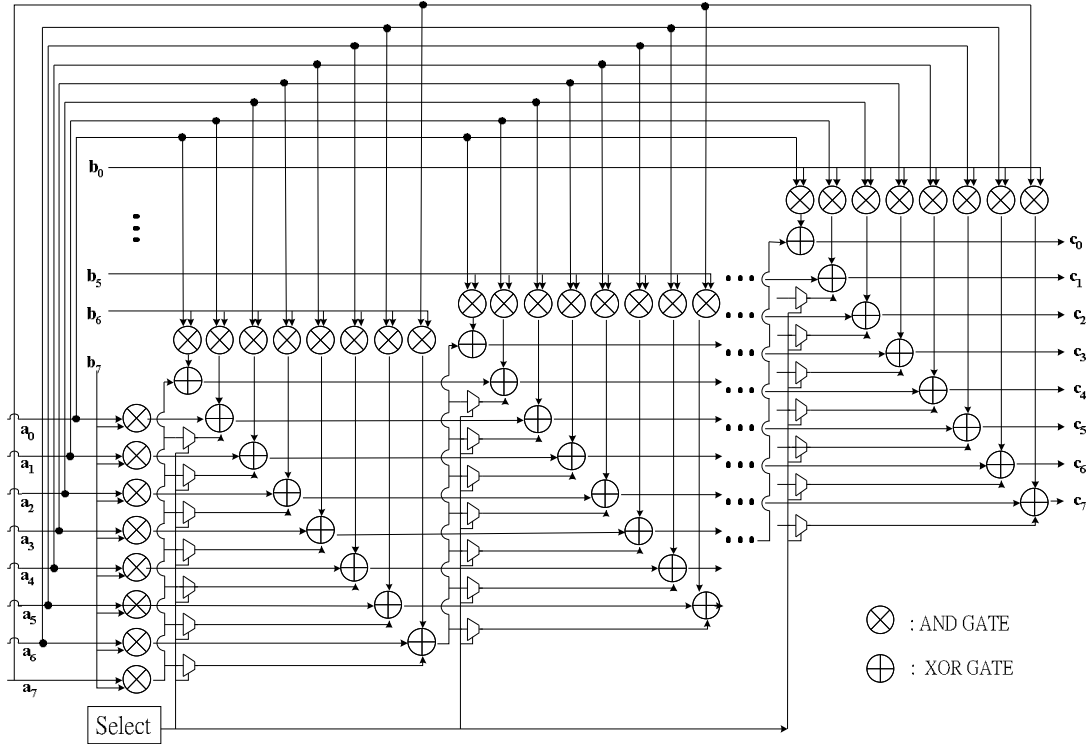


Figure 2. Implementation of an 8-bit multiplier.

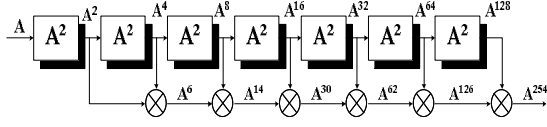


Figure 3. The finite field inverse circuit using square property.

According to the Figure 3, the inverse circuit has six multipliers in this kind of design. This inverse circuit is still a critical path in ByteSub module. However, the A^{254} can also be represented [4] as

$$A^{-1} = A^{2^8-1} = A^{254} = ((A^4) \cdot (A^3))^{32} \cdot ((A^3)^2 \cdot (A^3)^8) \quad (2)$$

Based on equation 2, the new inverse circuit architecture can be drawn and shown as Figure 4.

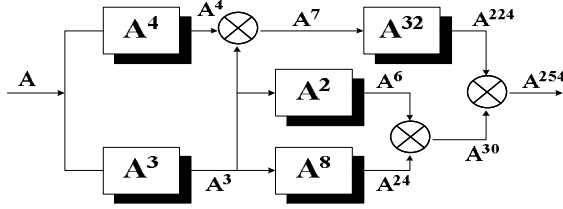


Figure 4. The new finite field inverse circuit.

As shown in Figure 4, the number of multipliers in this new finite field inverse circuit is reduced from 6 to 3 [4].

II. The Dynamic AES

The standard AES algorithm uses an irreducible binary polynomial of degree 8, $m(x) = x^8 + x^4 + x^3 + x + 1$, to perform the finite field multiplication in ByteSub transformation. According to R. Lidl and H. Niederreiter [7], we can apply the following theorem to count the number of irreducible polynomial [6] in $GF(2^m)$. In other words, we can use other irreducible polynomial over $GF(2^8)$ to construct different ByteSub transformations. Thus, the users can choose their own ByteSub transformation to perform the variant AES algorithm and increase the complexity.

Theorem : The number $N_q(n)$ of irreducible polynomial in $GF(2^m)$ of degree n is given by

$$N_q(n) = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d} \quad (3)$$

Möbius function μ :

$$m(n) = \begin{cases} 1 & \text{if } n = 1 \\ (-1)^k & \text{if } n \text{ is the product of } k \text{ distinct primes} \\ 0 & \text{if } n \text{ is divisible by the square of prime} \end{cases}$$

All of the 30 kinds of irreducible binary polynomial of degree 8 in $GF(2^8)$ were listed in Table 1.

We represent the 30 irreducible polynomials as a control bits set, then apply the corresponding control bits set (Table 1) to the proposed finite field multipliers (Figure 2) as the input signal. These finite field multipliers and inverse as shown in Figure 2 and 4 can construct the novel dynamic ByteSub generator with control bits set input signals.

4 The Implementation of Dynamic AES on FPGA and Cryptanalysis

In order to implement the proposed dynamic ByteSub generator, we apply the finite field inverse module and an Affine transformation module, which can be implemented directly on the bit-level. We also use a Field Programmable Gate Arrays (FPGA) chip to verify our novel hardware design. The Figure 5 gives the block diagram of this improved dynamic AES algorithm.

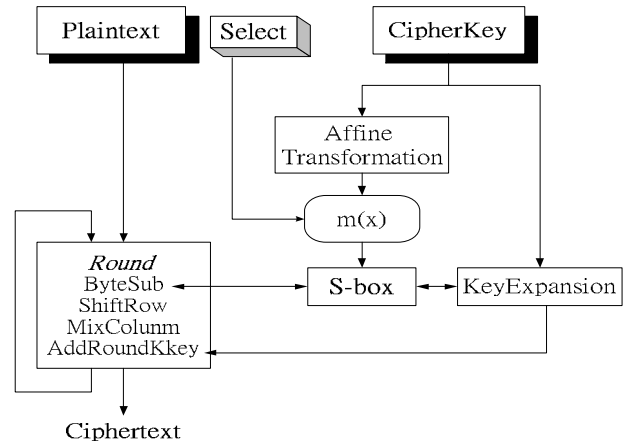


Figure 5. The block diagram of improved AES algorithm.

There is Electronic Code Book (ECB) mode and Cipher Block Chaining (CBC) mode [10,11] in this improved dynamic AES algorithm, with 128 bits data and key block. The $gmul_m$ signal, which uses the Table 1 control bits set as input bits, is a selector for 30 kinds of irreducible polynomials. The AES integral symbol is shown in Figure 6.

Table 1. All irreducible polynomial of degree 8 in GF(2⁸) and control component of ByteSub generator.

Index	Primitive	Irreducible Polynomial		Control Bit (binary)
0		$x^8+x^4+x^3+x+1$	1B	00000
1	*	$x^8+x^4+x^3+x^2+1$	1D	00001
2	*	$x^8+x^5+x^3+x+1$	2B	00010
3	*	$x^8+x^5+x^3+x^2+1$	2D	00011
4		$x^8+x^5+x^4+x^3+1$	39	00100
5		$x^8+x^5+x^4+x^3+x^2+x+1$	3F	00101
6	*	$x^8+x^6+x^3+x^2+1$	4D	00110
7	*	$x^8+x^6+x^4+x^3+x^2+x+1$	5F	00111
8	*	$x^8+x^6+x^5+x+1$	63	01000
9	*	$x^8+x^6+x^5+x^2+1$	65	01001
10	*	$x^8+x^6+x^5+x^3+1$	69	01010
11	*	$x^8+x^6+x^5+x^4+1$	71	01011
12		$x^8+x^6+x^5+x^4+x^2+x+1$	77	01100
13		$x^8+x^6+x^5+x^4+x^3+x+1$	7B	01101
14	*	$x^8+x^7+x^2+x+1$	87	01110
15		$x^8+x^7+x^3+x+1$	8B	01111
16	*	$x^8+x^7+x^3+x^2+1$	8D	10000
17		$x^8+x^7+x^4+x^3+x^2+x+1$	9F	10001
18		$x^8+x^7+x^5+x+1$	A3	10010
19	*	$x^8+x^7+x^5+x^3+1$	A9	10011
20		$x^8+x^7+x^5+x^4+1$	B1	10100
21		$x^8+x^7+x^5+x^4+x^3+x^2+1$	BD	10101
22	*	x^8+x^7+x+1	C3	10110
23	*	$x^8+x^7+x^6+x^3+x^2+x+1$	CF	10111
24		$x^8+x^7+x^6+x^4+x^2+x+1$	D7	11000
25		$x^8+x^7+x^6+x^4+x^3+x^2+1$	DD	11001
26	*	$x^8+x^7+x^6+x^5+x^2+x+1$	E7	11010
27		$x^8+x^7+x^6+x^5+x^4+x+1$	F3	11011
28	*	$x^8+x^7+x^6+x^5+x^4+x^2+1$	F5	11100
29		$x^8+x^7+x^6+x^5+x^4+x^3+1$	F9	11101

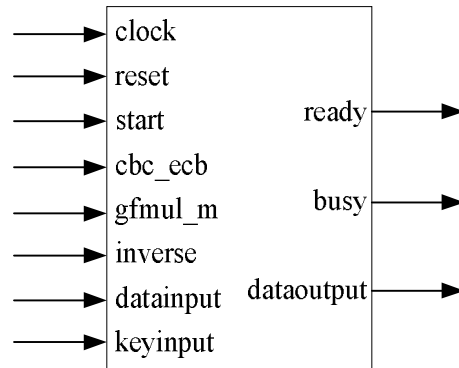


Figure 6. AES integral symbol.

The hardware description language we used in this paper is VHDL, and the synthesized device used is EP20K1500EBC652 of Altera's chip. There are 44218 logic cell elements in the simulations result and the performance achieves 57.2 MHz. Importantly, the proposed design does not need any memory bits elements. This feature can increase the security by preventing the exposure of S-Box data.

The Affine transformation has no impact on ByteSub transformation non-linearity properties. However, by proper chosen of Affine transformation, the S-Box will have a complex algebraic expression and the S-Box data will have no fixed points and opposite fixed point.

Cryptanalysis

To avoid the differential attack, the proposed novel dynamic AES preserve the original architecture and adds extra-modified parameter (gfmul_m signal) in ByteSub transformation. Furthermore, the systolic hardware design in the proposed one can be used to avoid power analysis attack and the timing attack.

5 Conclusions

In this paper, we implemented the dynamic AES on FPGA successfully. Using the proposed finite field inverse and multiplier modules, we add extra parameters in ByteSub transformation as the ByteSub generator. The users can choose one polynomial from the 30 kinds of irreducible polynomials freely, to generate their own version of S-Box tables. This proposed design can provide better performance and increase the security. In AES hardware implementation, we use the cell-based design to implement the proposed AES chip by using VHDL language. The future goal is to develop the dynamic modified Affine transformation and MixColumn transformation in subsequent study.

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